

CASS: Corrected Support Conditions and Operator Geometry

Technical companion to the author responses

27 September 2026

This companion gives complete proofs for the weighted objective and the actual hybrid intervention. It distinguishes the submitted operator from the additional contextual variant. These are conditional mathematical statements; the empirical comparison of operators and the geometry checks are separate evidence.

1 The weighted support condition

Let $U_t \in \mathbb{R}^{d \times r_t}$ have orthonormal columns, let $A = [U_1 \ \cdots \ U_T]$, and write the implemented objective as

$$\min_c \frac{1}{2} \|z - Ac\|_2^2 + \lambda \sum_{t=1}^T q_t \|c_t\|_2, \quad q_t = \sqrt{r_t} > 0. \quad (1)$$

Orthonormality holds within a block; distinct blocks need not be orthogonal. In the multi-layer construction, the joint blocks are block diagonal and r_t is their total number of columns.

Why the weights matter. An independently constructed counterexample makes the missing condition explicit. In $\mathbb{R}^5$, set

$$U_1 = [e_1, e_2, e_3, e_4], \quad U_2 = 0.6e_1 + 0.8e_5, \quad z = 1.01e_1, \quad \lambda = 0.01.$$

The true support is $S = \{1\}$, with coefficient $(1, 0, 0, 0)$ and noise $0.01e_1$. Maximum block coherence is 0.6, so the unweighted condition $|S| < (1 + 1/0.6)/2$ holds. However, $q_1 = 2$ and $q_2 = 1$. The optimum restricted to S is $(0.99, 0, 0, 0)$ and has residual $0.02e_1$. Thus

$$\left\| U_2^\top (z - U_1 \tilde{c}_1) \right\|_2 = 0.012 > \lambda q_2 = 0.01.$$

The inactive-block KKT condition fails. The full optimum has nonzero second block (numerically, $\hat{c}_{1,1} = 0.988125$ and $\hat{c}_2 = 0.003125$). The same failure occurs for every positive $\lambda < (1 + \epsilon)/2$ when $z = (1 + \epsilon)e_1$: changing only a constant relating λ to small noise does not repair the missing weights.

Proposition 1 (Weighted deterministic support bound). *Suppose $z = A_S c_S^0 + e$, $\|e\|_2 \leq \epsilon$, and A_S has full column rank. Define*

$$\begin{aligned} G &= A_S^\top A_S, & P &= A_S G^{-1} A_S^\top, \\ W_S &= \text{blockdiag}(q_t I_{r_t} : t \in S), \\ \theta_S &= \max_{j \notin S} \frac{1}{q_j} \sup_{\max_{t \in S} \|v_t\|_2 \leq 1} \left\| U_j^\top A_S G^{-1} W_S v \right\|_2, \\ \eta_S(e) &= \max_{j \notin S} \frac{\left\| U_j^\top (I - P) e \right\|_2}{q_j}. \end{aligned}$$

If $\theta_S < 1$ and $\lambda > \eta_S(e)/(1 - \theta_S)$, the unique optimum of (1) has support contained in S , and

$$\|\hat{c}_S - c_S^0\|_2 \leq \frac{\epsilon}{\sigma_{\min}(A_S)} + \frac{\lambda \sqrt{\sum_{t \in S} q_t^2}}{\sigma_{\min}(A_S)^2}. \quad (2)$$

Exact support follows if every true block norm exceeds the right side of (2). An $O(\epsilon)$ error conclusion additionally requires $\lambda = O(\epsilon)$ for a fixed dictionary.

Proof. Let $\tilde{c}$ minimize the objective restricted to S . Full column rank makes the restricted objective strictly convex, so its minimizer is unique. The active KKT equations give a block subgradient v , with $\|v_t\|_2 \leq 1$, such that

$$G(\tilde{c} - c_S^0) = A_S^\top e - \lambda W_S v.$$

Its residual therefore satisfies

$$r = z - A_S \tilde{c} = (I - P)e + \lambda A_S G^{-1} W_S v.$$

For every $j \notin S$,

$$\frac{\left\| U_j^\top r \right\|_2}{q_j} \leq \eta_S(e) + \lambda \theta_S < \lambda.$$

These strict inactive inequalities, together with the restricted KKT equations, show that extending $\tilde{c}$ by zeros gives a global optimum.

All optima have the same fitted value: otherwise strict convexity of squared loss in the fitted value, together with convexity of the penalty, would make their midpoint strictly better. Hence all optima have the same residual. A nonzero inactive block would require equality $\left\| U_j^\top r \right\|_2 = \lambda q_j$, contradicting the strict inequalities. All optima are thus zero outside S , and full column rank gives uniqueness. Finally,

$$\left\| G^{-1} A_S^\top \right\|_2 = \sigma_{\min}(A_S)^{-1}, \quad \left\| G^{-1} \right\|_2 = \sigma_{\min}(A_S)^{-2}, \quad \|W_S v\|_2 \leq \sqrt{\sum_{t \in S} q_t^2}$$

prove (2). Each block error is bounded by the total coefficient error, giving the stated sufficient minimum-amplitude condition. $\square$

This is a finite KKT argument for an established weighted group-selection condition, not a claim of new general sparse-recovery theory; see Bach (2008), Section 2.8. The support-capped finite-iteration solver is not automatically certified to use a noise-calibrated λ satisfying Proposition 1.

Corollary 1 (A conservative maximum-coherence condition). *Let $s = |S|$, $\mu = \max_{j \neq t} \|U_j^\top U_t\|_2$, and $R_S = \max_{t \in S} q_t / \min_{j \notin S} q_j$. If $(s - 1)\mu < 1$, then*

$$\theta_S \leq \frac{s\mu R_S}{1 - (s - 1)\mu}.$$

Consequently $\mu < 1/(sR_S + s - 1)$ implies $\theta_S < 1$.

Proof. In the block maximum norm, $\|G - I\| \leq (s - 1)\mu$. The Neumann series bounds $\|G^{-1}\| \leq [1 - (s - 1)\mu]^{-1}$. Moreover, $\|W_S v\|_{\text{block}, \infty} \leq \max_{t \in S} q_t$, while the off-support block row $U_j^\top A_S$ has induced norm at most $s\mu$. Combining these bounds and dividing by q_j proves the result. $\square$

Replacing R_S by the global maximum-to-minimum weight ratio gives a support-independent condition. Equal ranks recover $\mu < 1/(2s - 1)$. Also $\eta_S(e) \leq \epsilon / \min_{j \notin S} q_j$. A median coherence cannot replace the maximum. For joint block-diagonal dictionaries, each pair’s joint coherence is the maximum of its layerwise coherences.

2 Non-vacuity on the submitted dictionaries

We recomputed the five full dictionaries and all 160 LOTO dictionaries. The table gives full-dictionary values. “None” means the conservative global weighted corollary certifies no nonempty support, not that recovery is impossible.

Dictionary	Maximum μ	Global weighted s	Eligible two-block sets
Llama-3.1-8B	0.994555	None	10
Llama-3.2-3B	0.992069	1	1
Gemma-2-2B	0.988240	None	1
Qwen3-4B	0.993710	1	2
Qwen2.5-3B	0.993471	1	0

The final column checks all $\binom{32}{2} = 496$ supports in each dictionary using a sharper bound for θ_S . For $S = \{s_1, s_2\}$ and $j \notin S$, write

$$[M_1 \ M_2] = q_j^{-1} U_j^\top A_S G^{-1} W_S.$$

Duality and weighted Cauchy–Schwarz give, for any $t \in (0, 1)$,

$$\begin{aligned} \sup_{\|v_1\|, \|v_2\| \leq 1} \|M_1 v_1 + M_2 v_2\|_2 &= \max_{\|u\|_2=1} (\|M_1^\top u\|_2 + \|M_2^\top u\|_2) \\ &\leq \sqrt{\lambda_{\max} \left(\frac{M_1 M_1^\top}{t} + \frac{M_2 M_2^\top}{1-t} \right)}. \end{aligned}$$

Taking the minimum of evaluated upper bounds remains an upper bound; numerical scalar optimization need not find the exact supremum for this reasoning. The maximum bound over inactive blocks, plus a 10^{-6} numerical margin, determines eligibility. Llama-3.1’s park-country/word-length pair has bound 0.9417937318 before this margin and minimum Gram eigenvalue 0.6363702. None of the Llama-3.1 Compound constituent pairs is eligible. Gemma’s country-capital/word-length pair has bound 0.9978110662.

These geometric conditions do not establish the representation, noise, regularization or minimum-amplitude assumptions for observed task signatures. As implementation checks, three planted coefficient/noise instances for each of the 14 eligible pairs (42 total) recover their support and obey the coefficient bound under an explicitly admissible λ . They use the weighted objective at that prescribed λ , not the deployment support-cap selection rule.

3 A direct bound for the submitted hybrid

Fix the same hidden state h in two interventions, a gain $\gamma \geq 0$, a coefficient $\beta \geq 0$, and $\eta > 0$. For an orthogonal projector P , anchor μ , direction d and gate $g \in [0, 1]$, the submitted operator is

$$\alpha = \min\left(1, \frac{\beta \|(I - P)(h - \mu)\|_2}{\|h\|_2 + \eta}\right),$$

$$F(h) = h + \gamma(1 - g + g\alpha)d + g\alpha P(\mu - h).$$

Let starred quantities define a reference operator at the same h , and write $\delta_d = \|d - d^*\|_2$, $\delta_\mu = \|\mu - \mu^*\|_2$, $\delta_P = \|P - P^*\|_2$, $\delta_g = |g - g^*|$, $R = \|\mu^* - h\|_2$, $B = \delta_\mu + R\delta_P$, and $L_\alpha = \beta/(\|h\|_2 + \eta)$.

Proposition 2 (Fixed-state hybrid perturbation).

$$\|F(h) - F^*(h)\|_2 \leq \gamma\delta_d + B + (\gamma\|d^*\|_2 + R)(\delta_g + L_\alpha B). \quad (3)$$

Proof. Set $v = P(\mu - h)$ and $v^* = P^*(\mu^* - h)$. Projector norms are at most one, so $\|v - v^*\|_2 \leq B$ and $\|v^*\|_2 \leq R$. The same decomposition bounds the difference of $(I - P)(h - \mu)$ and its reference by B . Norm and scalar clipping are 1-Lipschitz; therefore $|\alpha - \alpha^*| \leq L_\alpha B$. Both $g\alpha$ and $1 - g + g\alpha$ lie in $[0, 1]$, and each changes by at most $\delta_g + |\alpha - \alpha^*|$. Bounding the direction and correction terms separately yields (3). $\square$

The implementation uses a shared gate in stacked layer space, whose stacked anchor $\bar{\mu}$ differs from the individual layer's anchor. For

$$g = \max\left(0, \frac{z^\top \bar{\mu}}{\|z\|_2 \|\bar{\mu}\|_2 + \epsilon_g}\right), \quad \epsilon_g \geq 0,$$

if both compared z norms are at least $a > 0$ and both anchor norms at least $b > 0$, then

$$\delta_g \leq \min(1, 2\|z - z^*\|_2/a + 2\|\bar{\mu} - \bar{\mu}^*\|_2/b).$$

Change z with $\bar{\mu}$ fixed, then $\bar{\mu}$ with z fixed. In each step the normalized map is $v/(\|v\|_2 + c)$ for a fixed $c \geq 0$, and its two-endpoint difference is at most $2\|v - v^*\|_2 / \min(\|v\|_2, \|v^*\|_2)$. The positive-part map is 1-Lipschitz. This proves the stated gate bound, including numerical denominator regularization.

For a fixed-context score whose gradient norm is at most K along the segment joining $F(h)$ and $F^*(h)$, the score difference is at most K times (3). Norm-threshold routing at τ is unchanged whenever $\|z - z^*\|_2 < \|\|z^*\|_2 - \tau\|$. These are explicit smooth-score and routing-margin conditions, not an unconditional generation bound.

4 Geometry of the additional contextual operator

The development-selected contextual variant has the different update

$$F(h) = h + \gamma d + aP(\mu - h - \gamma d), \quad a = cg \in [0, 1].$$

Here P is the selected-span projector, μ the weighted positive-context anchor, and d the direction from the demonstrations. The uncertainty coefficient c and cosine gate g are fixed after task adaptation. Set $b = h + \gamma d$. Then

$$(I - P)F(h) = (I - P)b, \quad PF(h) = (1 - a)Pb + aP\mu.$$

Thus the component orthogonal to the selected span is preserved exactly, and the selected component interpolates toward a composed contextual anchor. For a fixed task operator,

$$\begin{aligned} \|F(h) - F(h')\|_2^2 &= \|(I - P)(h - h')\|_2^2 + (1 - a)^2 \|P(h - h')\|_2^2 \\ &\leq \|h - h'\|_2^2. \end{aligned}$$

This follows from $P^2 = P$ and orthogonality of projected and complementary components. For two fixed operators, put $R = \|\mu^* - h - \gamma d^*\|_2$. Changing direction first, then anchor, then the coefficient/projector gives

$$\|F(h) - F^*(h)\|_2 \leq \gamma\delta_d + \delta_\mu + R(\delta_P + \delta_a).$$

The first change uses $\|I - aP\|_2 \leq 1$, the second uses $\|aP\|_2 \leq 1$, and the third uses $\|aP - a^*P^*\|_2 \leq \delta_P + \delta_a$.

The further weighted-skill correction uses $M = \sum_t w_t P_t$, $b_\mu = \sum_t w_t P_t \mu_t$, with $w_t \geq 0$ and $\sum_t w_t = 1$, and applies

$$F(h) = h + \gamma d + a\{b_\mu - M(h + \gamma d)\}.$$

Since $0 \preceq M \preceq I$ and $a \in [0, 1]$, $\|I - aM\|_2 \leq 1$: this fixed-state map is also nonexpansive. Rank-truncated weighted bases instead produce an orthogonal projector and obey the preceding projector argument. These facts explain what is composed; they do not assert sequential execution of named skills, nonexpansiveness of the subsequent transformer, or an increase in greedy exact-match accuracy.

Reproduction. The machine-readable evidence includes the original coherence audit, all support-specific bounds, 42 planted recovery checks, 1,200 submitted-hybrid perturbation checks, 1,000 contextual-geometry checks, and 480 weighted/rank nonexpansion checks. The last group also includes 120 explicit weighted-sum identity checks. These are numerical implementation checks rather than substitutes for the proofs above. All scripts, settings and raw results remain in the separate rebuttal and optimization directories.